

Open-YOLO 3D: Fast, Accurate Open-Vocabulary 3D Instance Segmentation

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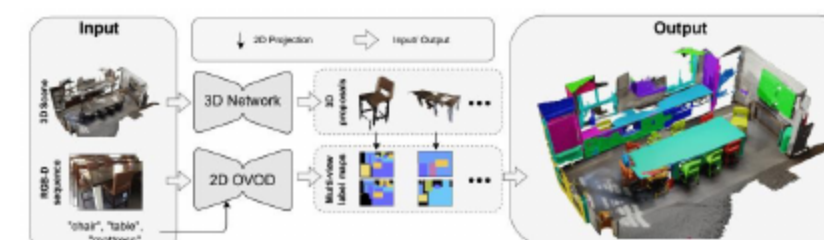
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Why Open-Vocabulary 3D?

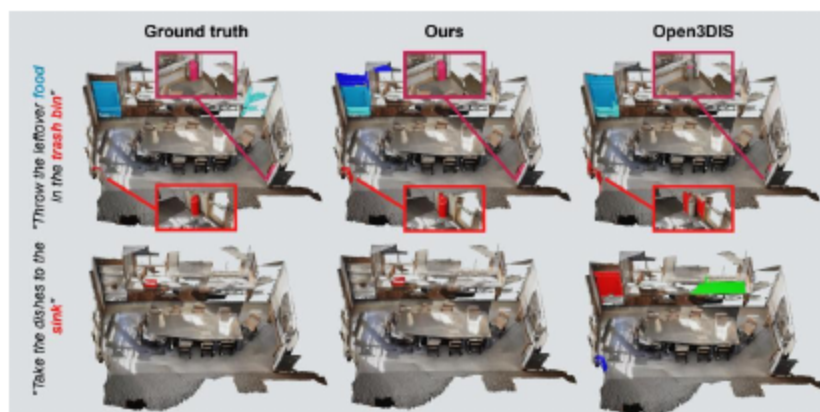
3 Key Challenges in 3D Instance Segmentation

- Computationally heavy models cause high-latency inference
- View-dependent appearance shifts hinder consistency
- Occlusions fragment instances, breaking 3D continuity
 - Sparse and noisy depth degrades geometry and recall
 - Imperfect 2D masks propagate errors into 3D labels
- Costly visibility checks and text-to-3D semantic linking

4 Open-YOLO 3D: Pipeline Overview



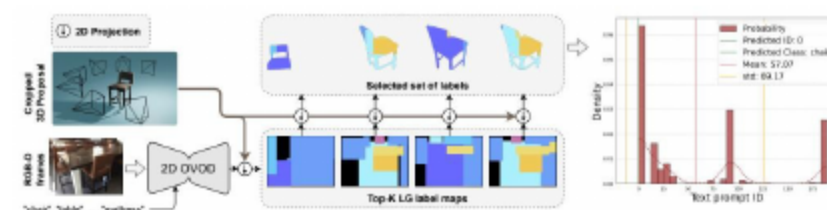
- End-to-end open-vocabulary 3D segmentation pipeline.
 - 3D proposal masks from a fast 3D segmentation net.
 - Labels assigned via per-frame 2D open-vocab detection.
- Visibility-aware aggregation fuses multi-view evidence.
 - Low-granularity label maps enable prompt-ID per mask.
 - Avoids heavy SAM/CLIP; adds confidence scoring.



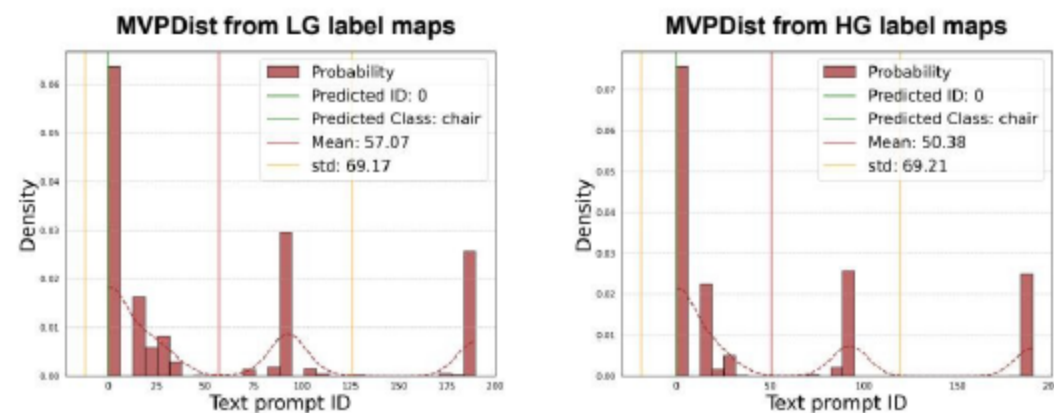
- Closed-vocab 3D segmentation misses unknowns.
 - OV 2D via CLIP recognizes beyond seen, yet 2D-only.
 - OV-3D (OpenMask3D, OVIR-3D) are slow feature lifts.
- Ours: fast 2D prompts; better GT match, stronger grounding.

Method: Open-YOLO 3D

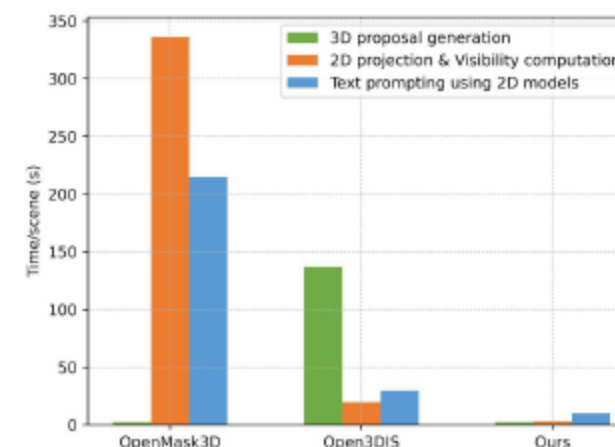
- 3D instance segmentation of point clouds from RGB-D input.
- Open-vocabulary: known and novel classes via text prompts.
- Assumes poses and camera parameters for reconstruction.
 - Input: RGB-D sequences paired with natural language.
 - Baseline uses 2D projections; drops separate segmentation stage.
- Uses 2D detection for mask visibility; per-frame labeling + fusion.



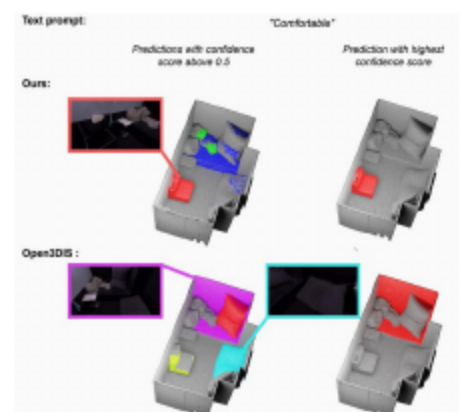
- 3D proposals via Mask3D with CNNs + transformers.
 - Project proposals to multi-view RGB-D frames.
 - Per-frame open-vocabulary detections assign labels.
- Build low-granularity label maps for efficiency.
 - Aggregate across views into consistent 3D instances.
 - Predict text prompt ID per mask via distributions.



- LG maps from 2D detections, not pixel masks
 - Lighter pipeline—faster label-map creation
 - MVPDist histograms: tighter, less noisy IDs
- Improves confidence calibration across prompts



- Batch visibility across frames accelerates checks
 - Reduced 2D projections via efficient batching
 - Fewer text prompts needed per scene or object
- Confidence score fuses class and mask evidence
 - Prunes low-value proposals using multiview overlap
 - Higher scene throughput; time per scene drops



- Scene-level, open-vocabulary: tighter masks, fewer FPs.
 - Beats prior mAP and speed on ScanNet200, Replica.
 - Well-calibrated confidence: class+mask fusion.
- Uses box overlap and multiview distribution for accuracy.

- Fast open-vocabulary 3D segmentation via efficient proposals.
- Integrates 2D detectors, reducing heavy 3D model demands.
- Robust low-/high-granularity label maps boost reliability.
 - Accelerated visibility reasoning improves segmentation speed.
 - Calibrated confidence scoring strengthens open-vocab trust.
- Strong qual/quant results; next: broader vocab, dynamic scenes.